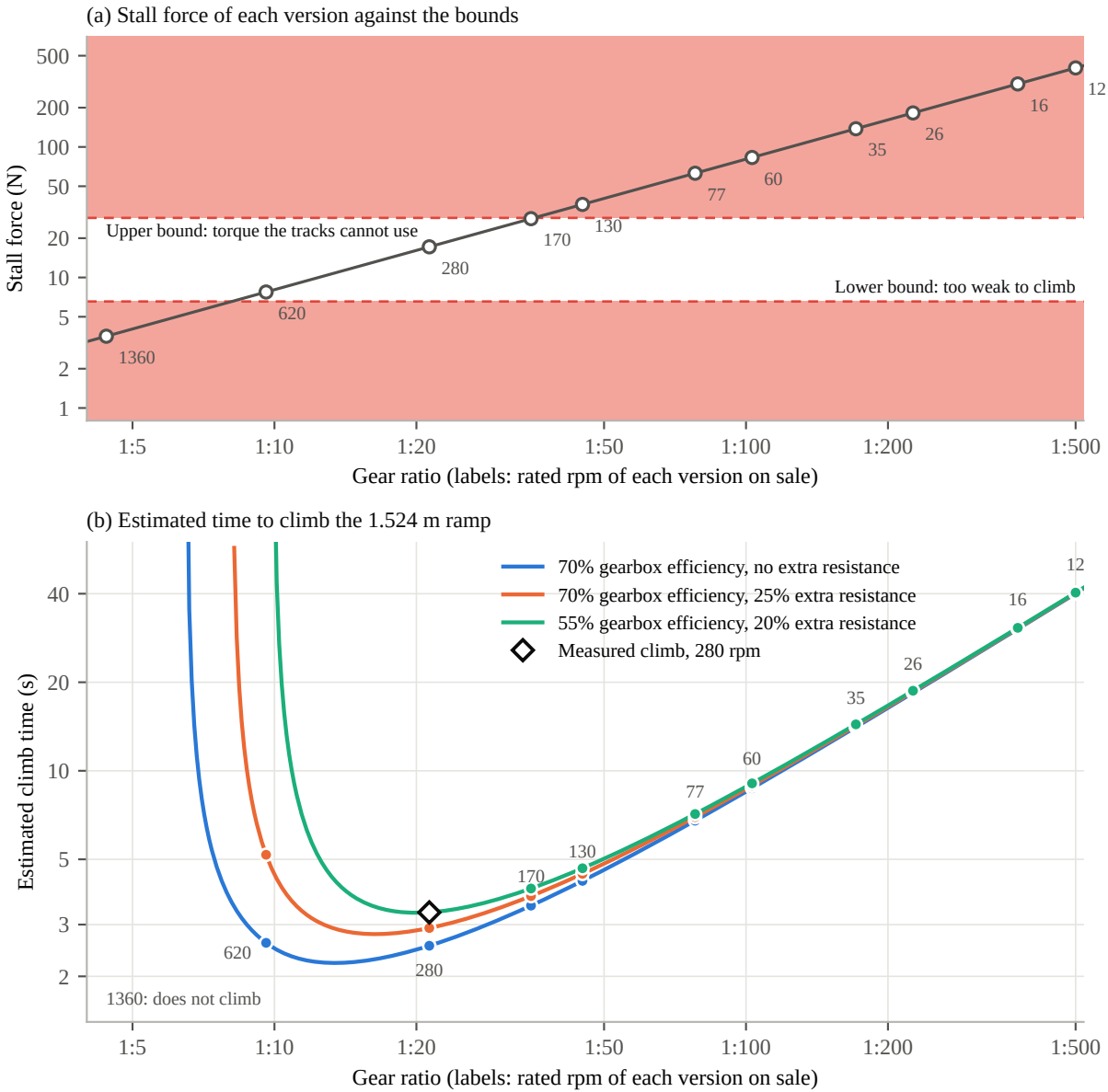
(a) Stall force of each version against the bounds
Stall force (N)
500
200
100
50
20
10
5
2
1
Upper bound: torque the tracks cannot use
Lower bound: too weak to climb
1360
620
280
170
130
77
60
35
26
16
12
1:5
1:10
1:20
1:50
1:100
1:200
1:500
Gear ratio (labels: rated rpm of each version on sale)
(b) Estimated time to climb the 1.524 m ramp
Estimated climb time (s)
40
20
10
5
3
2
70% gearbox efficiency, no extra resistance
70% gearbox efficiency, 25% extra resistance
55% gearbox efficiency, 20% extra resistance
Measured climb, 280 rpm
1360: does not climb
620
280
170
130
77
60
35
26
16
12
1:5
1:10
1:20
1:50
1:100
1:200
1:500
Gear ratio (labels: rated rpm of each version on sale)